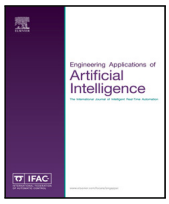
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Research paper



Enhancing online industrial quality index prediction with a general deep temporal feature extraction and incremental ensemble modeling framework

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ABSTRACT

Data-driven modeling methods for industrial quality index prediction often face the challenge of limited data representation. Using process variable snapshots from a single time step is insufficient for building high-performance soft sensors. Moreover, during online prediction, the performance of soft sensors is affected by diverse operating conditions and concept drift in industrial data streams, leading to performance degradation. To address these challenges, this paper proposes a Temporal Feature Extraction and Incremental Variational Bayesian Regression Ensemble (TFE-IVBRE) framework, which provides a general solution for various online industrial quality index prediction tasks. The TFE-IVBRE framework combines the temporal feature extraction capabilities of deep neural networks, the ability of ensemble learning to handle diverse operating conditions, and the online learning features of variational Bayesian models. An incremental update strategy is also incorporated to maintain consistent prediction performance. Experiments in the Debutanizer Column and Sulfur Recovery Unit scenarios show that the prediction performance of TFE-IVBRE significantly outperforms other static and online comparison models, with the effectiveness of its components validated through ablation studies. Finally, the overall model also demonstrates good robustness. These results offer valuable insights for the advancement of industrial soft sensor development.

1. Introduction

Soft sensor-based industrial quality index prediction plays a crucial role in modern industrial automation settings. It utilizes easily measurable process variables to indirectly infer and predict other variables that are difficult, time-consuming, or economically costly to measure directly. This facilitates a range of downstream tasks, such as monitoring, optimization and control. In recent years, data-driven industrial quality index prediction approaches have gained significant attention due to their modeling flexibility and lower reliance on expert knowledge. These methods have been successfully applied across various industries, including chemical engineering (Zhu et al., 2024), iron ore sintering (Yan et al., 2023), and cement production (Zhao et al., 2023, 2024b). Nevertheless, data-driven modeling methods for industrial quality index prediction still face several challenges that need to be addressed. Two of the main challenges currently encountered in this field can be summarized as:

Limitation of data representation for soft sensor modeling. Traditional methods for predicting industrial quality indices typically rely on simple-structured models that utilize process variable data from a single time step to represent the quality index (Kadlec et al., 2009). However, in most industrial settings, the relationship between the quality index and individual process variables is often weak and non-linear, especially considering the complex interactions and various types of noise exhibited by process variables. Moreover, given the dynamic nature of industrial processes and the time lags that exist between variables and quality indices, static data representations from individual time steps are often insufficient to fully characterize the overall operating conditions of industrial processes. As a result, relying solely on simple model structures to process snapshots of process variable data fails to capture the complexity and dynamic nature of industrial processes, thereby posing a significant challenge to the development of high-performance soft sensors.

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Inadequacy for diverse operating conditions and concept drift. Another issue closely related to industrial processes is that industrial devices often operate under various conditions that change over time. These conditions could involve shifts in process dynamics, equipment aging, or external disturbances, all of which require models capable of handling these different scenarios effectively. However, traditional models are typically designed and optimized for a specific set of conditions, but they struggle to generalize across a range of operating situations (Ge et al., 2023). As a result, predictive accuracy deteriorates when they are exposed to new, previously unseen conditions. This issue is further complicated by concept drift (Song et al., 2022), as it is primarily caused by the continuous changes in process characteristics and operating conditions (Kano and Fujiwara, 2013), and may also be triggered by some external factors or other elements not considered in the initial training data, ultimately resulting in the continuous change of the underlying data distribution over time. Therefore, to address these challenges, data-driven soft sensor models must be capable of dynamically adapting to continuously changing data, ensuring robust performance in all industrial environments (Chen and Liu, 2022).

The two major issues mentioned above have become bottlenecks in improving the performance of data-driven soft sensors, limiting their application in industrial quality index prediction. To tackle these challenges, this paper proposes a step-by-step approach to soft sensor modeling that is both versatile and scalable across different industrial scenarios. First, deep neural networks are applied to extract comprehensive temporal features from process variable data series, enhancing the model's input representation and improving its robustness. Then, a clustering-based ensemble learning method is applied to build an ensemble model with multiple regressors, allowing it to adapt to the diverse operating conditions typically encountered in industrial environments. Furthermore, to address concept drift, an incremental update strategy is proposed that combines minor updates for individual regressors with major updates for the entire ensemble model. This enables the model to continuously adapt to evolving data distributions in real-time.

As a result, a Temporal Feature Extraction and Incremental Variational Bayesian Regression Ensemble (TFE-IVBRE) framework is developed, which can maintain robust prediction performance across a wide range of industrial processes and varying operating conditions. The technical contributions of this framework are summarized below:

1. A Hierarchical Temporal Feature Extraction Autoencoder (HTFEAE) network is proposed for data representation enhancement. It leverages the strengths of self-attention mechanisms, Long Short-Term Memory (LSTM) networks, and Graph Convolutional Networks (GCN) to extract static, dynamic, and local features from time series data. These features are then refined using an autoencoder structure and a semi-supervised loss function to enhance their representational capability, improving the model's prediction performance and robustness.
2. A Variational Bayesian Regressor Ensemble (VBRE) model is proposed to handle multiple operating conditions. It clusters the feature dataset with MiniBatch K-Means and initializes a VBR sub-model for each cluster. During prediction, the right VBR sub-model is chosen based on the input feature-cluster centroid distance, which improves the soft sensor's ability to handle multiple operating conditions and complex feature distributions.
3. An incremental update strategy is proposed to tackle concept drift within the industrial data stream. This strategy enables the timely update of the corresponding VBR sub-models when new target labels arrive. Additionally, overall clustering and modeling operations are periodically re-executed to update the entire VBRE model, ensuring it adapts to the latest operating conditions at the same time.

The rest of the paper is organized as follows: Section 2 discusses the advancements and limitations of the data-driven industrial soft sensor modeling methods and online quality index prediction approaches. Section 3 provides a detailed introduction to the overall structure of the TFE-IVBRE and the technical details of its three key contributions. Section 4 presents the results of comprehensive experiments around the proposed TFE-IVBRE in two industrial scenarios: the Debutanizer Column (DC) and the Sulfur Recovery Unit (SRU). Finally, Section 5 concludes the paper and outlines the direction for future work.

2. Related work

2.1. Development of data-driven soft sensors

Early statistical learning-based regression methods, such as Principal Component Regression (PCR) (Mejdell and Skogestad, 1991), Partial Least Squares (PLS) (Mejdell and Skogestad, 1991; Kresta et al., 1994), and Support Vector Machine (SVM) regression (Yan et al., 2004), were applied in data-driven industrial soft sensor modeling. These methods can meet basic quality prediction needs, but have weak feature extraction capabilities and heavily rely on fixed mathematical transformations and manual intervention, making them unsuitable for complex industrial processes (Kadlec et al., 2009). Subsequently, the emergence of Multi-Layer Perceptrons (MLP) (Sun and Ge, 2021), a precursor to artificial neural networks, improved soft sensor performance by extracting rich feature combinations from process variable data, though it remained limited to snapshot-based modeling. It was not until the appearance of Convolutional Neural Networks (CNN) (Zhao et al., 2021) and Recurrent Neural Networks (RNN) (Li et al., 2019) that it became possible to build soft sensor models using time series data instead of data snapshots, which significantly enhances soft sensor models' data representation ability and prediction performance.

In recent years, increasingly complex network architectures, such as Graph Neural Networks (GCN) (Yan et al., 2024) and Transformers (Vaswani et al., 2017), have been applied to feature extraction tasks for further data representation enhancement. Additionally, in the field of industrial soft sensor modeling, the use of Autoencoders (AE) has gained significant attention. Some studies use stacked AEs to deeply extract features from data snapshots (Yuan et al., 2020c,a; Tian et al., 2024), while others employ AEs that include network structures like RNN and LSTM networks to capture compressed temporal features from time series data (Zhao et al., 2023; Yao et al., 2021). Furthermore, composite loss functions, such as semi-supervised learning loss (Zhao et al., 2024c), are employed during network training to maximize the utilization of available data. These advanced network architectures and training strategies have effectively addressed the limitations of data representation in soft sensor modeling.

However, several challenges persist in current industrial soft sensor modeling methods. First, industrial datasets are often smaller than those in other fields (Hao et al., 2021), making complex networks, architectures, and training processes less effective and prone to overfitting (Perera et al., 2023). Therefore, developing lightweight and general-purpose feature extraction networks is becoming increasingly important. Once effective features are extracted, research efforts should focus on how to best utilize them for downstream modeling and prediction tasks (Chen and Deng, 2023).

Moreover, for handling varying operating conditions, a potential solution is ensemble learning, where operating conditions are classified, and sub-models are created for each condition category (Liu et al., 2020). This enhances the model's ability to handle diverse scenarios but still faces challenges with continuous, long-term concept drift (Yeo et al., 2023). Periodic model retraining or updating can alleviate this issue. Nevertheless, frequent retraining consumes substantial computational resources, limiting its feasibility in practical applications, particularly for soft sensor models that are fully dependent on deep neural networks. Therefore, there is an urgent need for more efficient online prediction methods.

2.2. Online quality index prediction approaches

Compared to static soft sensor modeling and quality index prediction methods, online prediction focuses on dynamically updating model parameters within the industrial data stream to ensure continuous prediction accuracy. Early online prediction methods mainly involved recursive modeling (Wang et al., 2003) and moving window modeling (Liu et al., 2010). In recursive modeling, model parameters are updated in real-time, while moving window modeling periodically uses recent data within a time window to build the latest soft sensor model. However, these early methods have certain drawbacks: recursive modeling often requires significant computational resources, making it difficult to integrate with more complex models (Ditzler et al., 2015). In moving window modeling, the data range selection may result in the loss of important historical data, causing significant fluctuations in soft sensing performance.

As early online prediction methods developed, more flexible Just-In-Time Learning (JITL) approaches emerged and became widely used in industrial applications. For instance, Locally Weighted Regression (LWR) was proposed within a very early stage (Cleveland, 1979). It achieves local linearization modeling by assigning higher weights to neighboring data and has been successfully applied in industrial soft sensor modeling (Shigemori et al., 2011; Kim et al., 2013). Subsequently, more comprehensive JITL methods (Wang et al., 2022; Zhao et al., 2024a) have since been developed, which continue to attract significant attention (Yeo et al., 2023). The core idea of JITL is to dynamically select the most relevant data subset based on the current query sample to build a local model, thereby avoiding the computational burden of global modeling and enhancing the adaptability and efficiency of the model. This methodology gives birth to a series of more effective online index prediction methods, like incremental learning (Kingma and Welling, 2014) and lifelong learning techniques (Chen and Liu, 2022).

Incremental learning and lifelong learning typically begin by establishing a "knowledge base", which is updated within the industrial data stream to maintain real-time performance (Yao et al., 2021; Liu et al., 2024). These methods offer good scalability and can be effectively integrated with ensemble learning (Tian et al., 2019) and transfer learning (Li et al., 2023). However, as the knowledge base grows, it demands more computational resources, slowing down the modeling process (Grote-Ramm et al., 2023). Moreover, indiscriminate knowledge accumulation can lead to information redundancy and "catastrophic forgetting" issues (Kirkpatrick et al., 2017). To address these challenges, real-time knowledge base management strategies have been explored, including maintaining a fixed knowledge base size through First In First Out (FIFO) mechanisms and convex optimization (Grote-Ramm et al., 2023). However, these strategies often result in significant computational overhead, which greatly extends the resources needed for online prediction.

Consequently, recent advancements in online learning have focused on directly integrating incoming industrial data into model structures or parameters for faster updates. For example, the Broad Learning System (BLS) achieves knowledge integration by adding network nodes (Mou and Zhao, 2022). Another approach involves updating only a selected subset of model parameters directly using the input data, which allows for acquiring new knowledge while minimizing computational resource consumption (Mou et al., 2024). However, these neural network-based update strategies still require considerable computation time and thus may not fully meet the real-time demands of industrial applications. In this context, Bayesian regression offers a useful alternative. By combining new data with prior knowledge, it updates model parameters, helping prevent catastrophic forgetting and preserving important historical information (Yang et al., 2023). This ensures model stability while learning from new data (Zheng et al., 2024). Additionally, Bayesian regression naturally models uncertainty (Yuan et al., 2017), providing confidence intervals for predictions, which is critical for decision-making in real-time tasks. However, as a latent variable modeling approach, Bayesian regression may struggle with multi-condition and highly nonlinear industrial data. Thus, combining the feature extraction power of neural networks and the multi-condition handling capability of ensemble learning with Bayesian regression forms the central idea of this paper.

3. Methodology

The structure and workflow of the proposed TFE-IVBRE framework is shown in Fig. 1. In this framework, the HTFEAE network is first established using historical process variable data and their corresponding target labels. Additionally, the VBRE model is constructed by combining clustering techniques with the VBR algorithm. During the prediction task, the process variable series is constructed based on the newly inputted process variable data. The HTFEAE network is then employed to extract a comprehensive set of temporal features from the process variable series. These deep temporal features, along with the current process variable data (i.e., shallow features), are fed into the VBRE model. The model selects the optimal Bayesian regressor based on the distance between the input data and the cluster centers, and then outputs the corresponding predictions for the inputs. In the industrial data stream, as the delayed target labels corresponding to the process variable data arrive, the VBRE model is updated in real-time, allowing it to adapt to the latest operating conditions. The technical details of the TFE-IVBRE framework are described as follows.

3.1. Hierarchical temporal feature extraction

Temporal features extracted from the process variable series provide valuable insights into the temporal dynamics of the industrial process over a time period, thereby significantly enriching the input to the soft sensor. Given the unique characteristics of industrial process time series data, these temporal features can be broadly classified into three categories: static features, dynamic features, and local features. Static features capture the inherent properties of the variables within the time series, such as their average levels or distribution characteristics. Dynamic features reflect the time-dependent behaviors of the variables, including trends, periodic patterns, and other temporal dependencies. Finally, local features emphasize fine-grained fluctuations within the time series and the correlation patterns between different variables. The HTFEAE network, with its hierarchical structure, is designed to comprehensively extract the temporal features mentioned above.

The structure of the HTFEAE network is illustrated in Fig. 2. The input process variable series first passes through a self-attention mechanism (Vaswani et al., 2017) layer, which assigns weights to both the temporal and feature dimensions of the process variable series, thereby emphasizing the contributions of significant features and time steps:

$$\mathbf{X}_{\text{att}} = \text{softmax}\left(\frac{\mathbf{X}\mathbf{X}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{X} \tag{1}$$

Here, d_k represents the dimension of the input process variable series $\mathbf{X}$, and it is used to scale the attention weights. The process variable series after applying attention, which is denoted as $\mathbf{X}_{\text{att}}$, is then averaged over the time dimension to generate the static feature $\mathbf{v}_{\text{s}}$. To model the dynamic pattern of $\mathbf{X}_{\text{att}}$, a series of LSTM units are then applied:

$$\begin{cases} \mathbf{i}^t = \boldsymbol{\sigma}(\mathbf{W}_i[\mathbf{h}^{t-1}, \mathbf{x}_{\text{att}}^t] + \mathbf{b}_i) \\ \mathbf{f}^t = \boldsymbol{\sigma}(\mathbf{W}_f[\mathbf{h}^{t-1}, \mathbf{x}_{\text{att}}^t] + \mathbf{b}_f) \\ \mathbf{o}^t = \boldsymbol{\sigma}(\mathbf{W}_o[\mathbf{h}^{t-1}, \mathbf{x}_{\text{att}}^t] + \mathbf{b}_o) \\ \tilde{\mathbf{c}}^t = \tanh(\mathbf{W}_c[\mathbf{h}^{t-1}, \mathbf{x}_{\text{att}}^t] + \mathbf{b}_c) \\ \mathbf{c}^t = \mathbf{f}^t \odot \mathbf{c}^{t-1} + \mathbf{i}^t \odot \tilde{\mathbf{c}}^t \\ \mathbf{h}^t = \mathbf{o}^t \odot \tanh(\mathbf{c}^t) \end{cases} \tag{2}$$

LSTM effectively captures both long-term dependencies and short-term dynamics in time series through its long short-term memory units,

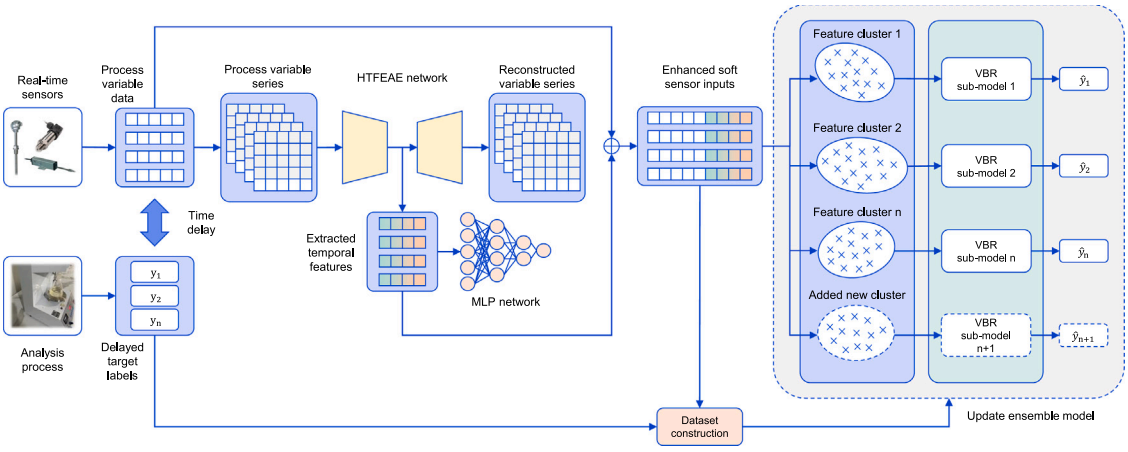


Fig. 1. The proposed TFE-IVBRE framework.

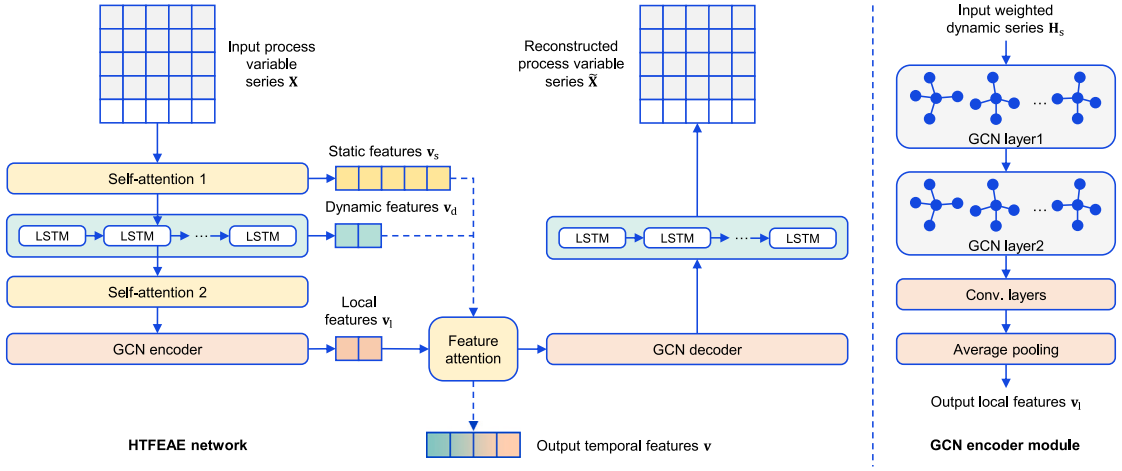


Fig. 2. Structure of the HTFEAE network.

thereby extracting rich dynamic features. The symbols $\mathbf{i}^t$, $\mathbf{f}^t$, $\mathbf{o}^t$, $\tilde{\mathbf{c}}^t$, $\mathbf{c}^t$, and $\mathbf{h}^t$ represent the value of input gate, forget gate, output gate, candidate cell gate, cell state, and hidden state. $\mathbf{h}^{t-1}$ and $\mathbf{c}^{t-1}$ stand for the previous hidden and cell states, respectively, carrying forward information across time steps. Additionally, the parameters $\mathbf{W}_i$, $\mathbf{W}_f$, $\mathbf{W}_o$, $\mathbf{W}_c$, $\mathbf{b}_i$, $\mathbf{b}_f$, $\mathbf{b}_o$, and $\mathbf{b}_c$ correspond to the weights and biases used in the respective operations. For the input series $\mathbf{X}_{\text{att}} = \{\mathbf{x}_{\text{att},1}, \mathbf{x}_{\text{att},2}, \ldots, \mathbf{x}_{\text{att},m}\}$ with the variable dimension m and length T, the LSTM units sequentially analyze each element, capturing its dynamic characteristics over time, and generate a dynamic series $\mathbf{H} = \{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_n\}$ which contains the hidden state information for each time step. According to the operation of the LSTM unit as described in Eq. (2), the data at the last time step of $\mathbf{H}$, i.e., $\mathbf{H}^T$, can be regarded as the dynamic features $\mathbf{v}_{\text{d}}$ extracted from the process variable series.

To effectively capture local features, a hybrid approach combining GCN and CNN is employed, implemented through a GCN encoder module. As illustrated in Fig. 2, another self-attention layer is first applied to the dynamic series to generate a weighted dynamic series $\mathbf{H}_{\text{att}}$, and a graph structure is constructed based on the output $\mathbf{H}_{\text{att}}$. Specifically, for $\mathbf{H}_{\text{att}} = \{\mathbf{h}_{\text{att},1}, \mathbf{h}_{\text{att},2}, \ldots, \mathbf{h}_{\text{att},n}\}$, each time series $\mathbf{h}_{\text{att},i}$ corresponds to a node v_i in the graph. The Euclidean distance between two series $\mathbf{h}_{\text{att},i}$, $\mathbf{h}_{\text{att},j}$ is defined as the weight w_{ij} of edge e_{ij} between v_i and v_j:

$$w_{ij} = d(v_i, v_j) = \sqrt{\sum_{t=1}^{T} (v_i^t - v_j^t)^2} \tag{3}$$

Where T is the length of the time series, and v_i^t denotes the value of the ith variable at time point t. Based on the obtained nodes and edges, a symmetric adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is constructed, whose elements a_{ij} are the edge weights w_{ij}. Using the GCN layers, the graph structure and the matrix $\mathbf{A}$ is processed to aggregate information from neighboring nodes and edges, outputting an enriched set of node representations series $\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_o\}$, with each $\mathbf{z}_i$ embodying both the structural connectivity and temporal correlations intrinsic to the input $\mathbf{H}_{\text{att}}$. To capture fine-grained temporal variations within $\mathbf{Z}$, a convolutional layer is introduced. This layer applies convolutional kernels of varying sizes to perform localized weighted summation operations on $\mathbf{Z}$, utilizing an activation function to enhance the nonlinearity of the feature extraction process:

$$\mathbf{O}^t = \sigma\left(\sum_{i=0}^{k-1} \mathbf{w}_i \mathbf{Z}^{t \cdot i}\right) \tag{4}$$

Where k denotes the size of the convolution kernels, and $\mathbf{w}$ represents their corresponding weights. A pooling layer is then applied to adaptively summarize the feature maps produced by the convolutional layer, yielding the local features $\mathbf{v}_{\text{l}}$ of the input time series.

Finally, a feature attention layer is introduced to weight $\mathbf{v}_{\text{g}}$, $\mathbf{v}_{\text{d}}$, and $\mathbf{v}_{\text{l}}$. This process integrates a comprehensive set of temporal features $\mathbf{v}$, which serve as the output of the network. The current process variable data $\mathbf{x} = \{x_1, x_2, \ldots, x_m\}$ and the temporal features $\mathbf{v}$ are combined, which serve as the feature input $\mathbf{f} = \{\mathbf{x}, \mathbf{v}\}$ to the proposed VBRE model.

To learn representative temporal features from the input process variable series, a symmetric network structure is constructed as a

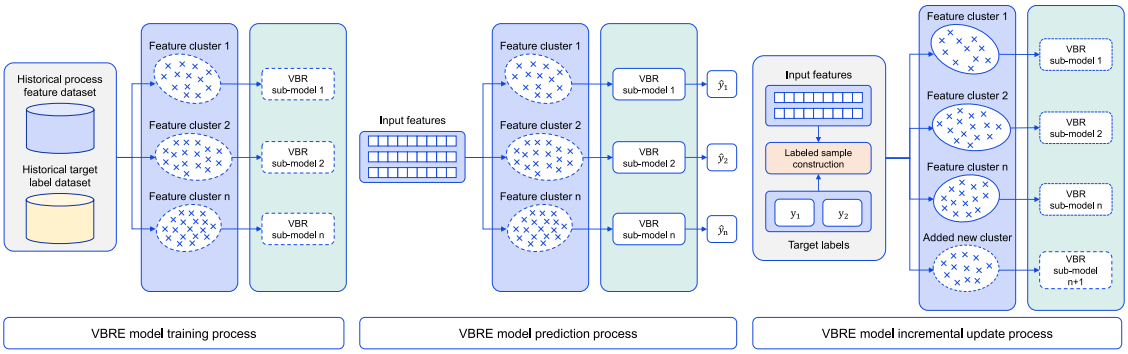


Fig. 3. The training, prediction, and incremental update process of the VBRE model. The dashed border indicates an initialized object, and the dash–dot border indicates an updated one.

decoder, which reconstructs the process variable series $\mathbf{X}$ from the temporal features $\mathbf{v}$ output by the feature attention layer. This enables unsupervised learning, which accounts for the name HTFEAE. Additionally, to establish a connection between temporal feature extraction (TFE) and the prediction task, a Multilayer Perceptron (MLP) network is introduced and combined with the HTFEAE network to form the HTFEAE-MLP network. This integration allows the output $\mathbf{v}$ to be used to predict the corresponding target labels of the input $\mathbf{X}$, thereby introducing a supervised loss term into the training of the HTFEAE network. Based on this design, the training loss function of the HTFEAE network can be expressed as:

$$\mathcal{L} = \frac{\lambda_1}{N \times T \times D} \sum_{n=1}^{N} \sum_{t=1}^{T} \sum_{d=1}^{D} (\mathbf{X}_{n,d}^{t} - \tilde{\mathbf{X}}_{n,d}^{t})^2 + \frac{\lambda_2}{N} \sum_{n=1}^{N} (y_n - \hat{y}_n)^2 \tag{5}$$

This training loss consists of two terms, weighted by hyperparameters λ_1 and λ_2, respectively. The first term is the reconstruction loss, which calculates the mean squared error (MSE) between the input process variable series $\mathbf{X}$ and the reconstructed series $\tilde{\mathbf{X}}$ across all samples (N), time steps (T), and features (D). The second term is the supervised loss, which measures the MSE between the predicted target label $\hat{Y}_b$ and the true label y_b for each sample. By training the HTFEAE network with this composite semi-supervised loss function, the network can be better guided to extract key temporal information from the process variable series to support industrial prediction tasks.

3.2. Ensemble model construction

As mentioned before, individual soft sensor models may struggle to capture the differences across multiple operating conditions in industrial processes. In contrast, ensemble learning strategies are particularly effective at addressing the complexities associated with multi-condition industrial processes. Based on this consideration, the VBRE model is developed. The operations of the proposed VBRE model can be divided into three processes: training, prediction, and incremental update, as illustrated in Fig. 3. To build the VBRE model, the HTFEAE-MLP model is first trained using the historical process variable time series data $\mathbf{X}$ and their corresponding target labels y. Based on this, the HTFEAE network is used to generate the historical $\mathbf{v}$ dataset from the historical $\mathbf{X}$, which is combined with the historical $\mathbf{x}$ to form the comprehensive feature dataset $\mathbf{F} = \{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_N\}$. Next, the MiniBatch K-Means algorithm (Sculley, 2010) is introduced to cluster the feature vectors in dataset $\mathbf{F}$. Despite the existence of many other mature clustering algorithms, such as Gaussian Mixture Models (GMM) (Carreira-Perpinan, 2000) algorithm and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) (Schubert et al., 2017) algorithm, DBSCAN algorithm is first ruled out due to its lack of stability and controllability. Secondly, considering the computational efficiency of clustering, MiniBatch K-Means is selected for its clear advantages on industrial datasets (Patel and Kushwaha, 2020), making it suitable to be applied in the proposed framework. Specifically, in the MiniBatch K-Means algorithm, K centroids $\mu_1, \mu_2, \ldots, \mu_K$ are first initialized by randomly selecting K data points from $\mathbf{F}$. At each iteration, a mini-batch of data points is randomly selected from the dataset. The distance between each data point in the mini-batch and each centroid is calculated using the squared Euclidean distance $d_{ik} = \|\mathbf{f}_i - \mu_k\|^2$ where $\mathbf{f}_i$ represents the ith data point, and μ_k denotes the centroid of the kth cluster. Each data point is then assigned to the closest centroid by selecting the minimum distance:

$$k_i = \arg\min_k d_{ik} \tag{6}$$

After assigning the data points to their respective clusters, the centroids are updated by computing the mean of the data points assigned to each cluster. The update rule for the centroids is given by:

$$\mu_k \leftarrow \mu_k + \frac{1}{n_b} \sum_{i=1}^{n_b} (\mathbf{f}_i - \mu_k) \tag{7}$$

Where n_b is the size of the mini-batch. This process is repeated until the centroids converge, i.e., the change in centroids between iterations is below a certain threshold or a maximum number of iterations is reached. Once the clustering process is complete, the feature vectors in the historical $\mathbf{F}$ dataset are partitioned into K clusters:

$$\mathbf{F}_1 = \{\mathbf{f}_i \mid k_i = 1\}, \quad \mathbf{F}_2 = \{\mathbf{f}_i \mid k_i = 2\}, \ldots, \quad \mathbf{F}_K = \{\mathbf{f}_i \mid k_i = K\} \tag{8}$$

Where $\mathbf{F}_k$ represents the set of data points assigned to the kth cluster, and k_i is the cluster index assigned to each data point $\mathbf{f}_i$. By using MiniBatch K-Means, the computational efficiency is significantly improved, making it feasible to handle large-scale historical data during the initialization of the VBRE model (Béjar Alonso, 2013). Furthermore, the MiniBatch K-Means algorithm's efficiency is also crucial for the incremental update of the VBRE model in industrial data streams.

Based on the clustered sub-feature datasets $\mathbf{F}_k$ obtained from MiniBatch K-Means, along with their corresponding target label datasets $\mathbf{Y}_k$, a VBR sub-model is initialized for each cluster. Specifically, for each cluster k, the VBR sub-model aims to learn a probabilistic mapping between the feature set $\mathbf{F}_k$ and the corresponding target labels $\mathbf{Y}_k$ (Broderick et al., 2013). The initialization process of the VBR sub-model can be expressed as follows:

Given data points $\mathbf{F}_k = \{\mathbf{f}_i \mid k_i = k\}$ and their corresponding target labels $\mathbf{Y}_k = \{y_i \mid k_i = k\}$, the objective is to learn the posterior distribution of the regression weights $\mathbf{w}_k$ given the features $\mathbf{F}_k$ and the labels $\mathbf{Y}_k$. The likelihood of the weights can be modeled as:

$$p(\mathbf{Y}_k \mid \mathbf{F}_k, \mathbf{w}_k) = \mathcal{N}(\mathbf{Y}_k \mid \mathbf{F}_k \mathbf{w}_k, \sigma^2 I) \tag{9}$$

Where $\mathcal{N}(\cdot)$ denotes the Gaussian distribution, $\mathbf{w}_k$ represents the regression weights for cluster k, and σ^2 is the noise variance. To perform variational Bayesian inference, the posterior distribution of the weights

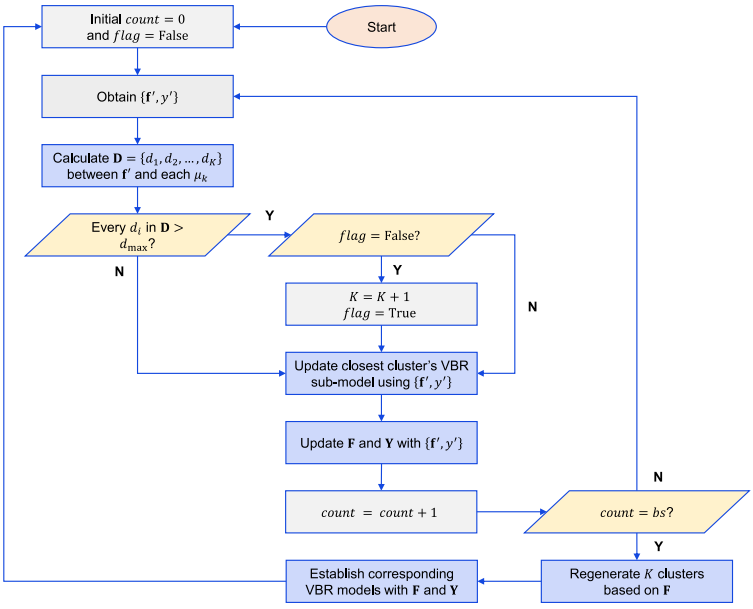


Fig. 4. Flowchart of VBRE model's incremental update strategy.

is approximated by a variational distribution $q(\mathbf{w}_k)$. The evidence lower bound (ELBO) for the variational distribution is given by:

$$\mathcal{L}(q) = \mathbb{E}_{q(\mathbf{w}_k)}[\log p(\mathbf{Y}_k \mid \mathbf{F}_k, \mathbf{w}_k)] - D_{KL}[q(\mathbf{w}_k) \parallel p(\mathbf{w}_k)] \tag{10}$$

Where $D_{KL}[\cdot \parallel \cdot]$ denotes the Kullback–Leibler divergence, and $p(\mathbf{w}_k)$ is the prior distribution of the weights, typically assumed to be a Gaussian distribution with zero mean and a fixed variance.

Maximizing the ELBO concerning the variational parameters provides the best approximation of the posterior distribution $q(\mathbf{w}_k)$, which can then be used to make predictions. The initialized VBR sub-model for each cluster k is thus characterized by the learned posterior distribution of $\mathbf{w}_k$, enabling it to be updated within the data streams.

When making predictions with the trained VBRE model, the corresponding feature vector $\mathbf{f}'$ is first obtained from the newly input time series $\mathbf{X}'$. Then, the distance between $\mathbf{f}'$ and each centroid μ_k is calculated. Finally, the VBR sub-model corresponding to the centroid μ_k with the smallest distance to $\mathbf{f}'$ is selected to predict the corresponding target $\hat{y}$, where the prediction $\hat{y}$ is made by the VBR sub-model for cluster k as:

$$\hat{y} = \mathbf{f}'\mathbf{w}_k \tag{11}$$

The proposed ensemble modeling method divides the process features into distinct subsets corresponding to different patterns through clustering, with each subset being modeled independently. Compared to the utilization of a single soft sensor model, this approach better captures the characteristics of multiple operating conditions in industrial processes, thereby improving the efficiency and accuracy of industrial prediction tasks.

3.3. Incremental update strategy

Considering the aforementioned concept drift issues encountered by the constructed VBRE model in industrial data streams, which lead to a gradual deterioration of the model's prediction performance over time, an incremental update strategy is proposed. This strategy updates the VBRE model as soon as the target label y' corresponding to the newly input process feature vector $\mathbf{f}'$ is obtained. The flowchart of this strategy is illustrated in Fig. 4. Overall, the updates to the VBRE model can be classified into two modes: minor update and major update. The minor update involves updating the regression weights $\mathbf{w}_k$ of the individual VBR sub-models, and it occurs each time a complete pair of $\{\mathbf{f}', y'\}$ is received. In contrast, the major update involves updating the entire clustering pattern as well as $\mathbf{w}_k$ of all the VBR sub-models. This update happens only after accumulating a certain batch size of $\{\mathbf{f}', y'\}$. The designed batch size, denoted as bs, can be customized according to the specific characteristics of the modeled industrial process and the prediction task.

Between two consecutive major updates, the parameters *count* and *flag* are first initialized. The *count* represents the number of $\{\mathbf{f}', y'\}$ pairs accumulated during this period, while *flag* indicates whether an additional cluster is needed during the major update that follows the end of this period. Next, a $\{\mathbf{f}', y'\}$ pair is obtained, and the distance $D = \{d_1, d_2, \ldots, d_K\}$ between the feature vector $\mathbf{f}'$ and each cluster center μ_k is calculated. If the distance between $\mathbf{f}'$ and every μ_k exceeds a preset threshold $d_{\max}$, it suggests that the current clustering pattern can no longer meet the requirements of the feature distribution, and consideration should be given to increasing the number of clusters.

Notably, to prevent the rapid increase in the number of clusters that leads to redundancy in VBR sub-models, the *flag* is used to limit the addition of at most one cluster per major update period. Subsequently, the $\{\mathbf{f}', y'\}$ pair is used to update the regression weights $\mathbf{w}_k$ of the VBR sub-model corresponding to the nearest cluster center to $\mathbf{f}'$, while also updating the feature set $\mathbf{F}$ and target label set $\mathbf{Y}$. This process is repeated until the number of $\{\mathbf{f}', y'\}$ pairs accumulated in this period reaches *count*, at which point a major update is executed. During the major update, K clusters are generated based on the latest $\mathbf{F}$, and for each cluster, K VBR sub-models are reconstructed using $\mathbf{F}$ and $\mathbf{Y}$, thereby performing an overall update of the entire VBRE model.

In industrial data streams, continuously cycling through the minor and major update processes ensures that the constructed VBRE model can adapt to changes in data distribution. Specifically, considering the characteristics of the VBRE model, the designed update strategy not only responds in real-time to small-scale data drifts through frequent minor updates but also adjusts to large-scale concept changes periodically via major updates. This adaptability is crucial for maintaining the real-time prediction performance of the VBRE model.

Moreover, the integration of multiple VBR sub-models enables the VBRE model to handle complex data distributions by combining multiple different Gaussian distributions, significantly increasing its robustness. This approach shares similarities with the logic of Gaussian Mixture Regression (GMR) models (Yuan et al., 2014). However, compared to GMR, the proposed update mechanism allows the model to dynamically adjust its complexity according to actual needs, offering a more

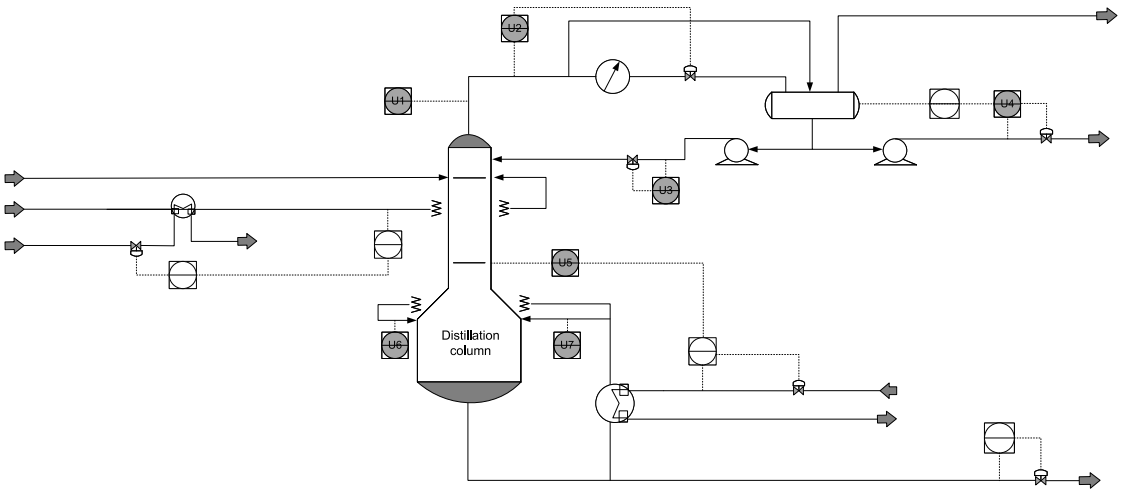


Fig. 5. The flowchart and variable selection of DC process (Wang et al., 2022; Fortuna et al., 2007).

refined functionality. For instance, when changes occur in the feature space and existing clusters cannot adequately describe new situations, new clusters are promptly added to enhance the model's expressive power. Conversely, when the data characteristics are relatively stable, the number of clusters will not be increased. This means that the number of sub-models contained in the VBRE model remains unchanged, thereby avoiding unnecessary complexities to reduce computational resource consumption and simplify model maintenance.

4. Experiment

4.1. Dataset introduction

To assess the general applicability and performance of the proposed TFE-IVBRE model in real-world industrial scenarios, a series of experiments is designed and conducted using two open-source process industry datasets: the DC dataset and the SRU dataset. These datasets can be found here (Fortuna et al., 2007).

The main function of the DC is to remove C_4 (butane) and C_3 (propane) from the naphtha stream. This process maximizes the C_5 (stabilized gasoline) content at the distillation column top and minimizes the C_4 content within the bottom product, which is crucial for subsequent refining processes and the production of other products. Overall, the variation in the C_4 content of the bottom product is relatively smooth, though sudden fluctuations can be significant. Due to the delay caused by the distance between the measurement equipment and the DC device, using a soft sensor to estimate the C_4 content has become an effective solution. As shown in Fig. 5, in the DC scenario, the selected modeling variables include top temperature (U1), top pressure (U2), reflux flow (U3), flow to the next process (U4), 6th tray temperature (U5), and the average of two bottom temperatures (U6, U7), with the soft sensor modeling target being the aforementioned C_4 content.

On the other hand, the main purpose of the SRU is to process industrial tail gases containing hydrogen sulfide (H_2S), such as monoethanolamine (MEA) gas and sour water stripping (SWS) gas, and to recover valuable elemental sulfur through the Claus reaction (Sendt and Haynes, 2005). This process is more complex than the reactions occurring in the DC scenario. Specifically, the MEA gas mainly contains H_2S, while the SWS gas contains both H_2S and ammonia (NH_3). These gases enter two separate combustion chambers and undergo partial combustion and catalytic conversion under the supply of controlled air. During the combustion process, H_2S is partially oxidized to form sulfur dioxide (SO_2), which then reacts with the unreacted H_2S to produce elemental sulfur (S) and water (H_2O) within the reaction furnace. This series of reactions can be expressed as:

$$\begin{aligned} 2H_2S + 3O_2 &\rightarrow 2SO_2 + 2H_2O \\ 2H_2S + SO_2 &\rightarrow 3S + 2H_2O \end{aligned} \tag{12}$$

After the Claus reaction, the remaining tail gas still contains small amounts of H_2S and SO_2, which will be further treated in the tail gas treatment unit. Most of the time, the concentrations of H_2S and SO_2 in the tail gas remain relatively stable, but occasional spikes occur, which is different from the variations in the C_4 content within the debutanizer (DC) process. Establishing soft sensors for the real-time measurement of H_2S and SO_2 is of great significance, as it helps to simultaneously guide the air supply in the combustion chamber and the adjustment of the tail gas treatment unit parameters. As shown in Fig. 6, in the SRU process, the modeling variables for the soft sensor include the flow rate of MEA gas (U1), air flow (U2), secondary air flow (U3), flow rate of SWS gas (U4), and air flow in the SWS zone (U5). Regarding the modeling target, previous studies on soft sensor modeling in SRU systems suggest that predicting H_2S concentration is generally more challenging than SO_2 concentration. Moreover, in practical applications, monitoring H_2S is often of higher priority due to its critical impact on process control. Therefore, in the context of SRU, H_2S is selected as the representative modeling target to provide a more in-depth and focused analysis.

4.2. Experimental settings

For the training and testing of the TFE-IVBRE model on both datasets, the first 80% of the data from each dataset is used for training, while the remaining 20% is reserved for online testing. Specifically, during the training of the HTFEAE-MLP network, the last 25% of the training data from each dataset is set aside as a validation set for its hyperparameter tuning. During the experiments, the modeling and online testing process of the TFE-IVBRE model is as follows:

Training: For the TFE-IVBRE modeling, the training sets of both the DC and SRU datasets are first used to construct the corresponding HTFEAE-MLP network models. Then, using the trained HTFEAE network, data series are constructed based on the process variable data from the training set, and the corresponding temporal features are generated. These features are subsequently added to the training set. Finally, the extended training set is used to initialize the clusters and their corresponding VBR sub-models, completing the construction of the TFE-IVBRE model.

Online testing: For the online testing of the TFE-IVBRE model, the process is conducted based on the testing dataset: Each time a new process variable data sample is encountered, a data series is first

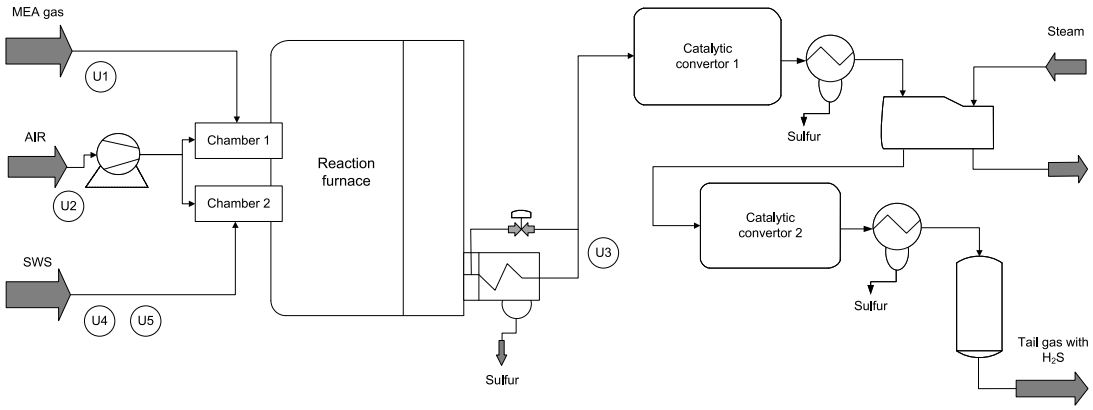


Fig. 6. The flowchart and variable selection of SRU process (Wang et al., 2022; Fortuna et al., 2007).

Table 1
Key parameters for TFE-IVBRE modeling and online testing.

Parameter	Value	Parameter	Value
Time series length	30	λ_1	0.5
$\mathbf{v}_s$ number	7 (DC), 5 (SRU)	λ_2	0.5
$\mathbf{v}_d$ number	6	Initial cluster number K	3
$\mathbf{v}_l$ number	4	Prior precision	1
Learning rate	10e-3 (DC), 10e-5 (SRU)	Noise precision	1
Training epoch	50	bs	30
Batch size	36	$d_{\max}$	4

constructed from this sample. Then, the trained HTFEAE network is used to extract temporal features from the constructed data series. Finally, based on the concatenation of the process variable data and the temporal features, the distance to each cluster centroid is calculated, and the appropriate VBR sub-model is selected for prediction. Meanwhile, the corresponding target label is used to execute the designed incremental update strategy. Specifically, to reflect the time delay in obtaining the target label, a delay of 30 samples between the input process variable data and the target label is defined in both the DC and SRU scenarios. This delay prevents the use of the target label corresponding to the current process features for just-in-time minor updates to the VBRE model, which aligns with the real-world data update mechanisms and the prediction process in industrial settings.

The key parameters used to build and conduct online testing of the TFE-IVBRE model on the DC and SRU datasets are provided in Table 1. To highlight the differences between the two datasets, efforts are made to keep the parameters for building and testing the TFE-IVBRE model as consistent as possible across both datasets.

4.3. Static model performance comparison

To explore the performance differences between soft sensor models that use process data snapshots and those that use process variable data series, and to understand the upper limit of static models' prediction performance, a series of soft sensors based on static modeling methods are constructed and tested on the DC and SRU datasets. These modeling methods include PLS regression (Lin et al., 2007), MLP (Pani et al., 2013), Extreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016), Ensemble Support Vector Machine (ESVM) regression (Liu et al., 2020), and Semi-Supervised Stacked Autoencoders (SSAE) (Yuan et al., 2020a), all of which use data snapshots for modeling, as well as Multivariate Time Series Analysis and CNN (MVTS-CNN) (Zhao et al., 2021), LSTM (Tian et al., 2023), Transformer (Fang et al., 2024), and Dynamic Feature Extraction and Transferring Model (DFETM) (Yao et al., 2021), which all use process variable data series. Additionally, the intermediate model HTFEAE-MLP constructed during the TFE-IVBRE modeling process is also tested to demonstrate the TFE and prediction capabilities of the HTFEAE network. The parameters for all replicated methods are tuned according to the literature to achieve their optimal performance.

To ensure that the testing results are statistically significant, for all network-based methods, 20 repeated modeling-testing processes are conducted, and the average of the testing results is taken as the final result. The predictions of the models constructed by all methods on the DC and SRU testing dataset are visualized, as shown in Figs. 7 and 8. Moreover, four performance metrics are calculated to evaluate the prediction results and their corresponding true values: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Spearman's Rank Correlation Coefficient (SPCC), and Coefficient of Determination (R^2). Among them, MAE and RMSE reflect the magnitude of the error between the predicted and true values, providing an intuitive measure of the prediction accuracy. SPCC reflects the correlation between the predicted and true values in terms of their trends, thus measuring whether the model's prediction direction is correct, while R^2 serves as a comprehensive evaluation metric:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{\mathbf{Y}}_i - \mathbf{Y}_i| \tag{13}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{\mathbf{Y}}_i - \mathbf{Y}_i)^2} \tag{14}$$

$$\mathrm{SPCC} = 1 - \frac{6\sum_{i=1}^{N} d_i^2}{N(N^2-1)}, d_i = r(\mathbf{Y}_i) - r(\hat{\mathbf{Y}}_i) \tag{15}$$

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{N}(\mathbf{Y}_i - \hat{\mathbf{Y}}_i)^2}{\sum_{i=1}^{N}(\mathbf{Y}_i - \bar{\mathbf{Y}})^2} \tag{16}$$

Here, N represents the total number of true labels $\mathbf{Y}$ (or predicted values $\hat{\mathbf{Y}}$), with $\bar{\mathbf{Y}}$ indicating the average of $\mathbf{Y}$, and $r(\cdot)$ referring to the rank of a sample within the entire dataset. Finally, the modeling time for all models is recorded. For the network-based modeling results, the average of their performance metrics is taken. The performance metrics of all model testing results are listed in Table 2.

The testing results on the DC dataset suggest that early modeling approaches based on data snapshots, such as PLS, MLP, XGBoost, and ESVM, generally show suboptimal performance. These models face difficulties in adapting to the sharp increase in target labels during the

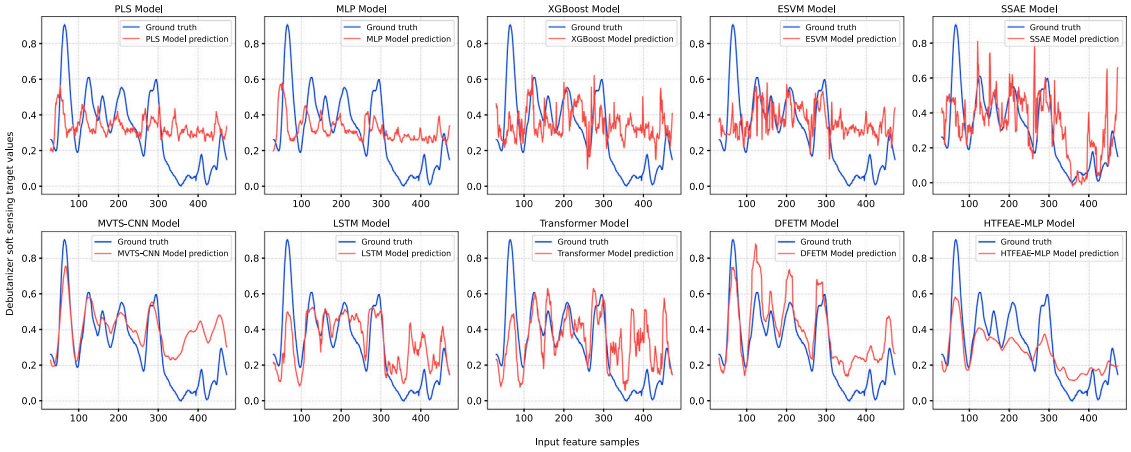


Fig. 7. Visualization of the static model testing results on the DC testing dataset.

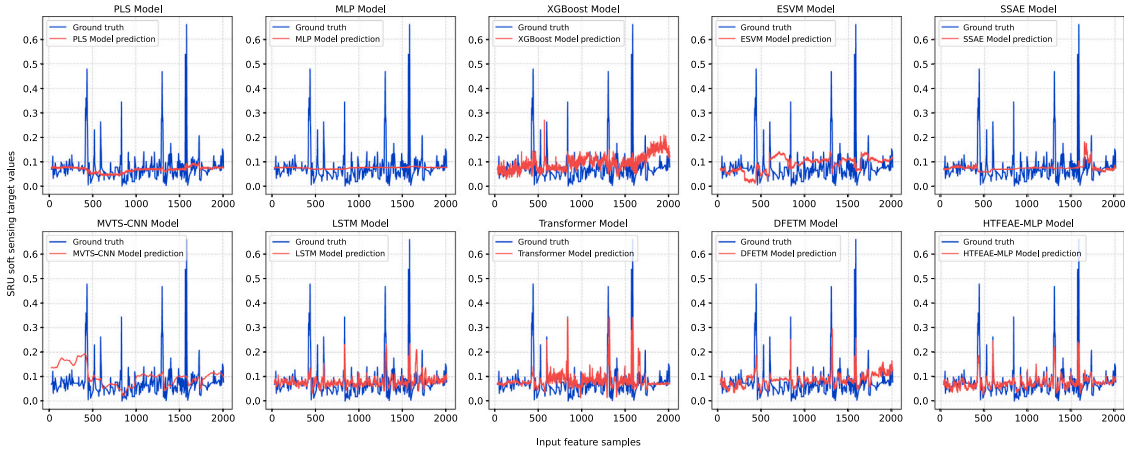


Fig. 8. Visualization of the static model testing results on the SRU testing dataset.

Table 2
Performance metrics of static models on the DC and SRU testing datasets.

DC dataset						SRU dataset					
Model	MAE	RMSE	SPCC	R^2	Time (s)	Model	MAE	RMSE	SPCC	R^2	Time (s)
PLS	0.1641	0.1986	0.4134	0.0979	0.0100	PLS	0.0281	0.0564	0.3265	0.0163	0.0070
MLP	0.1629	0.1965	0.4619	0.1169	5.5855	MLP	0.03	0.0566	0.2537	0.0092	22.999
XGBoost	0.1705	0.2089	0.2609	0.0016	24.009	XGBoost	0.047	0.0687	0.0438	−0.4574	60.484
ESVM	0.157	0.1986	0.4522	0.0975	9.0521	ESVM	0.0464	0.0671	0.0289	−0.3887	40.775
SSAE	0.1218	0.1705	0.6779	0.3353	8.7768	SSAE	0.0309	0.0579	0.0964	−0.0356	37.411
MVTS-CNN	0.1281	0.1702	0.7674	0.3376	7.5852	MVTS-CNN	0.0486	0.0661	0.3906	−0.3483	31.607
LSTM	0.1198	0.154	0.7418	0.4573	8.8468	LSTM	0.0287	0.0492	0.4405	0.2514	25.680
Transformer	0.139	0.1819	0.5796	0.2434	23.433	Transformer	0.0304	0.0494	0.3757	0.2453	89.345
DFETM	0.1194	0.1368	0.8932	0.5722	75.763	DFETM	0.0319	0.0469	0.5553	0.3219	327.12
HTFEAE-MLP	**0.0986**	**0.119**	**0.9382**	**0.6760**	**16.941**	**HTFEAE-MLP**	**0.0214**	**0.0392**	**0.6927**	**0.5249**	**166.37**

first 100 data points and the subsequent steep decline after the 300th data point. SSAE, which enhances the data representation extraction from data snapshots through AE (Yuan et al., 2020a), performs slightly better, but its predictions remain highly volatile. This variability reflects the inherent challenge of obtaining stable modeling features from data snapshots. In contrast, modeling methods that utilize data series, such as MVTS-CNN, LSTM, and Transformer, significantly improve the prediction performance, demonstrating the value of TFE in soft sensor modeling. Notably, although Transformer utilizes a more advanced network architecture which requires longer training times, its performance is weaker than that of MVTS-CNN and LSTM. This indicates that excessively complex networks may be prone to overfitting, which can reduce the effectiveness of soft sensor modeling in industrial scenarios. The DFETM model, which combines the TFE ability of AE with the regression capability of XGBoost (Yao et al., 2021), outperforms the three baseline time-series models. Visualization of the results confirms that its outputs are more reasonable. Finally, the HTFEAE-MLP model, which is designed to focus on diverse TFE, outperforms all comparison models in terms of prediction performance. Its modeling time is also at a relatively reasonable level.

The findings from the SRU scenario generally align with those on the DC dataset, but there are also some differences that provide new insights. Compared to the DC process, the SRU process has more complex mechanisms and a larger dataset, which undoubtedly increases the difficulty of soft sensor modeling. The performance of modeling methods based on data snapshots remains poor, and even the performance of enhanced feature extraction methods like SSAE struggles on the SRU dataset. The increase in dataset size seems to intensify

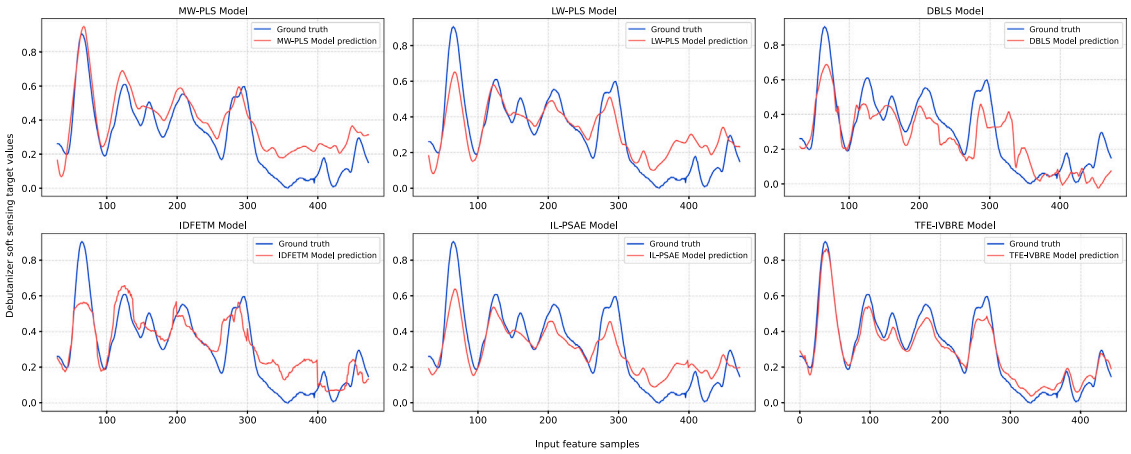


Fig. 9. Visualization of the online prediction results on the DC testing dataset.

the limitations of these models, making them more easily overfitted. Among the three baseline time-series feature extraction methods, the LSTM and Transformer models are still able to maintain a certain level of prediction performance, while the performance of MVTS-CNN shows significant degradation. This may be because the pure CNN structure can only capture relatively short-term temporal relationships in the data series (Yuan et al., 2020b), making it less effective than LSTM and Transformer in capturing long-term dependencies. Therefore, MVTS-CNN is not suitable as a TFE network for soft sensor models. Since the AE component of the DFETM model consists of an LSTM network and fully connected layers, it still exhibits relatively good performance. Finally, although its modeling time is extended, the HTFEAE-MLP model still achieves the best prediction performance on the SRU dataset, highlighting its generalizability in TFE. However, due to the static nature of HTFEAE-MLP, it encounters difficulties in predicting many finer details, highlighting the limitations of static modeling approaches to some extent.

4.4. Online prediction results analysis

Furthermore, the performance of the proposed TFE-IVBRE is evaluated on the DC and SRU datasets. At the same time, several online modeling methods are introduced for comparison with this approach. These methods include the Moving Window PLS (MW-PLS) regression (Liu et al., 2010) based on the moving window modeling technique, the Local Weighted PLS (LW-PLS) regression (Kim et al., 2013) utilizing JITL, the Deep Broad Learning System (DBLS) (Mou and Zhao, 2022), as well as the Incremental DFETM (IDFETM) approach (Yao et al., 2021) and the Incremental Learning-Parallel Stacked Autoencoder (IL-PSAE) method (Mou et al., 2024), which are both developed based on incremental learning strategies.

For all comparison methods, parameter tuning is conducted based on their respective literature. Specifically, the model updates for these online prediction methods also follow the target label delay rule mentioned in Section 4.2. In addition, according to the results in Section 4.3, the use of data snapshots inevitably has a negative impact on modeling performance. Therefore, for all online modeling methods using two-dimensional data, the temporal average of the data series, rather than snapshots, is uniformly used as input, which improves their comparability with methods that directly use data series. For each method, 20 rounds of online modeling and prediction are conducted, and the average result is taken to avoid contingency. The visualized prediction results for each method are shown in Figs. 9 and 10, with the performance metrics of their prediction results listed in Table 3.

On the DC dataset, even the simplest MW-PLS method exhibits relatively good results. The LW-PLS method shows similar performance to MW-PLS but benefits from a faster locally weighted update process (Cleveland, 1979), enhancing its efficiency in online modeling. While the performance metrics of DBLS's predictions are comparable to those of the first two methods, visualizing its predictions reveals a notable performance decline after the 200th sample point, where it fails to accurately and promptly capture the variations in the true C_4 values. Furthermore, during repeated experiments, the online modeling performance of DBLS proves to be highly unstable, which can be attributed to the fact that feature nodes in BLS are derived from the hidden layer features extracted by AE (Mou and Zhao, 2022). The increase and update of nodes actually amplify the uncertainty introduced by AE training. Given that modeling stability is a crucial evaluation criterion for soft sensor modeling, this instability significantly undermines its practical utility. By incorporating the incremental update strategy, the IDFETM method enhances its prediction accuracy, but its prediction results become more volatile. Additionally, because the knowledge base update process in this early work does not account for the selective inclusion of knowledge, operational time increases substantially. In contrast, IL-PSAE improves both prediction performance and operational efficiency by selectively sampling the knowledge base for direct updates to network parameters (Mou et al., 2024). However, IL-PSAE's inability to extract temporal features prevents it from reaching optimal performance. Finally, the proposed TFE-IVBRE method strengthens the online modeling input by utilizing the temporal features, and through a Bayesian regression strategy, maximizes the retention of key information from historical data, which all leads to a significant improvement in prediction performance. Moreover, the focus on updates limited to clustering and Bayesian methods eliminates time-consuming network updates, greatly reducing the time needed for online modeling.

In contrast to the DC dataset, the SRU dataset not only involves the complex mechanisms discussed earlier but also features a much larger scale of data samples. The SRU dataset consists of over 10000 samples, with more than 2000 samples within the testing dataset. These differences create an excellent opportunity to evaluate whether the replicated online modeling methods can maintain robust performance over time in complex industrial environments. The results from the online prediction tests on the SRU dataset highlight clear and significant performance differences across the online modeling methods.

The MW-PLS method's performance in the SRU scenario is relatively low, primarily due to the inherent information loss associated with the moving window technique. The LW-PLS method mitigates this issue, yet its prediction performance is limited by the incapability of the PLS approach in complex scenarios. The DBLS exhibits some predictive ability, however, due to the inherent instability of BLS in online modeling, its predictions are even less accurate than those of the LW-PLS method. Specifically, when compared to the results from the DC

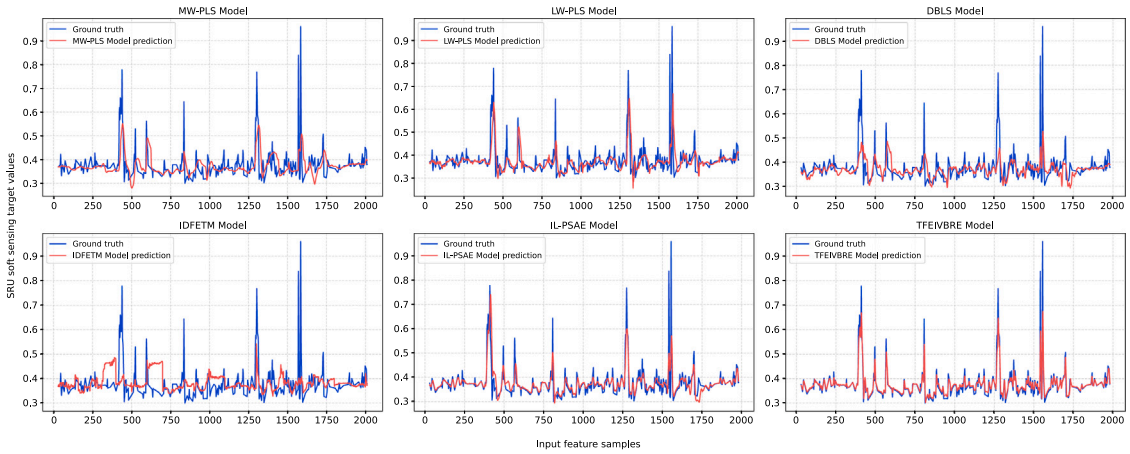


Fig. 10. Visualization of the online prediction results on the SRU testing dataset.

Table 3
Performance metrics of online modeling methods on the DC and SRU testing datasets.

DC dataset						SRU dataset					
Model	MAE	RMSE	SPCC	R^2	Time (s)	Model	MAE	RMSE	SPCC	R^2	Time (s)
MW-PLS	0.1011	0.1176	0.9271	0.6834	1.6831	MW-PLS	0.0305	0.0517	0.2773	0.1741	6.8111
LW-PLS	0.0911	0.1168	0.8587	0.688	0.9970	LW-PLS	0.0247	0.0437	0.5164	0.4096	5.9282
DBLS	0.0957	0.1241	0.7944	0.6479	4.1196	DBLS	0.0251	0.0465	0.5638	0.3418	5.3390
IDFETM	0.084	0.1116	0.8719	0.7151	414.82	IDFETM	0.0384	0.0616	0.2048	−0.1723	4044.1
IL-PSAE	0.0838	0.1054	0.9100	0.7460	7.7509	IL-PSAE	0.0157	0.0332	0.8091	0.6637	32.943
TFE-IVBRE	**0.0395**	**0.0461**	**0.9953**	**0.9513**	**1.0613**	**TFE-IVBRE**	**0.0097**	**0.0210**	**0.9506**	**0.8652**	**5.1841**

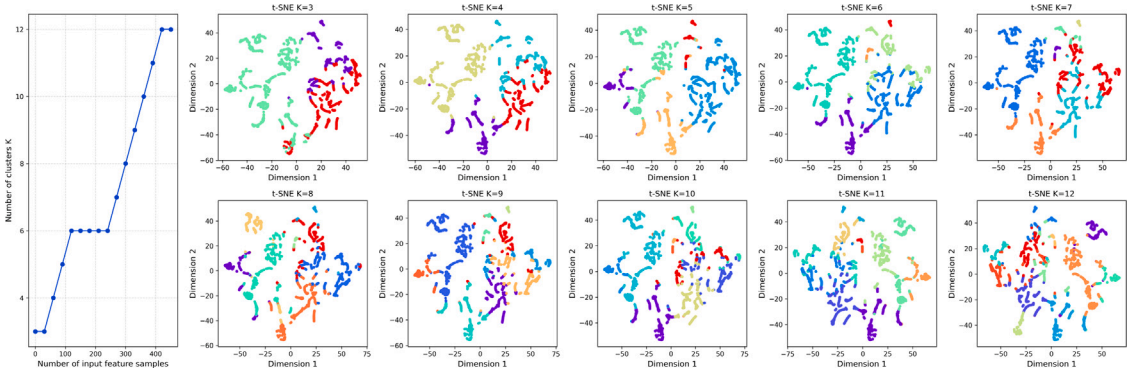


Fig. 11. Variation in cluster number and shape during TFE-IVBRE's incremental update on the DC dataset.

scenario, the online prediction performance of IDFETM exhibits a significant deterioration. By comparison with the DFETM predictions, the catastrophic forgetting phenomenon is clearly presented. Additionally, the runtime of IDFETM has increased substantially, which highlights that the uncontrolled expansion of the knowledge base not only results in a noticeable decline in prediction performance during long-term online modeling tasks but also imposes a detrimental effect on its operational efficiency. However, with the implementation of a well-managed knowledge base and model update strategy, IL-PSAE continues to demonstrate robust performance. Finally, the TFE-IVBRE method consistently achieves superior online prediction performance and operational efficiency on the SRU dataset, demonstrating its exceptional performance and practicality.

4.5. Incremental update process visualization

In order to intuitively understand the incremental update process of the TFE-IVBRE model, the overall incremental update process in the DC and SRU scenarios is plotted. This includes the variation in the number of clusters as the TFE-IVBRE model processes input feature samples, as well as the t-SNE visualizations of the feature dataset **F** under different cluster numbers. First, the visualization results of TFE-IVBRE on the DC dataset are shown in Fig. 11. From the curve of cluster number variation on the left, it can be observed that as new feature samples are input, the number of clusters in the TFE-IVBRE model gradually increases. After the 100th data point, the number of clusters stabilizes, indicating that the current cluster number is sufficient to handle the new operating conditions. However, after the 250th data point, the number of clusters increases again, suggesting that a new operating condition is identified, thereby prompting the expansion of clusters. The t-SNE scatter plots on the right demonstrate that, under different cluster numbers, the clustering results effectively distinguish various distinct feature patterns. These observations preliminarily validate the rationality of the designed incremental update algorithm.

Although the TFE-IVBRE's incremental update process is reasonable in the DC scenario, the relatively small size of the DC dataset naturally raises a concern: will TFE-IVBRE generate an excessively large number of clusters during its prolonged operation? To address this, the incremental update process of TFE-IVBRE on the SRU dataset is also visualized, with results exhibited in Fig. 12.

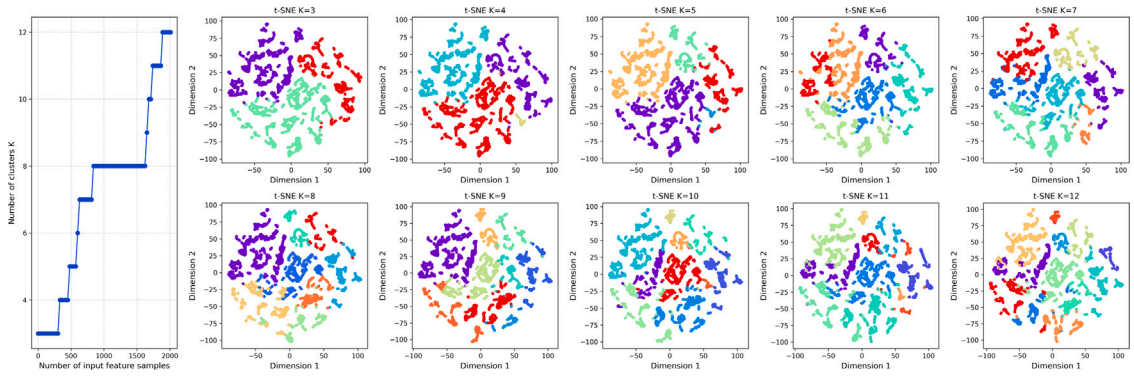


Fig. 12. Variation in cluster number and shape during TFE-IVBRE's incremental update on the SRU dataset.

Table 4
Performance metrics of TFE network ablation study methods on the DC and SRU testing datasets.

DC dataset					SRU dataset				
Model	MAE	RMSE	SPCC	R^2	Model	MAE	RMSE	SPCC	R^2
CNNAE-IVBRE	0.0610	0.0763	0.9845	0.8703	CNNAE-IVBRE	0.0148	0.0288	0.9157	0.7470
GCNAE-IVBRE	0.0611	0.0719	0.9910	0.8849	GCNAE-IVBRE	0.0143	0.0288	0.9225	0.7467
LSTMAE-IVBRE	0.0535	0.0655	0.9883	0.9043	LSTMAE-IVBRE	0.0126	0.0252	0.9365	0.8067
TransformerAE-IVBRE	0.0516	0.0610	0.9829	0.9170	TransformerAE-IVBRE	0.0134	0.0257	0.9290	0.7994
TFE-IVBRE	**0.0395**	**0.0461**	**0.9953**	**0.9513**	**TFE-IVBRE**	**0.0097**	**0.0210**	**0.9506**	**0.8652**

Despite the SRU testing dataset being approximately four times larger than the DC dataset, the final number of clusters still stabilizes at 12. This indicates that the proposed incremental update strategy operates strictly based on operating condition changes rather than the size of datasets, preventing an excessive increase in cluster numbers due to dataset size, thus effectively alleviating the aforementioned concern. Additionally, in the SRU scenario, the number of clusters in TFE-IVBRE increases uniformly during the input of feature samples and remains constant across the middle 1000 data points. This suggests that the operating conditions in the SRU testing dataset exhibit an intermittent pattern, differing from those in the DC dataset. The clustering visualizations show that TFE-IVBRE effectively groups the feature set $\mathbf{F}$ across different cluster numbers. The results from the SRU scenario further provide intuitive validation of the TFE-IVBRE's effectiveness.

4.6. Ablation study

A comprehensive series of ablation studies is conducted to further assess the design of each component in the proposed TFE-IVBRE model. First, to verify the suitability of the HTFEAE network structure for TFE, AE networks based on CNN, GCN, LSTM, and Transformer structures are constructed. These networks keep the attention module unchanged but replace other hybrid structures in HTFEAE for TFE tasks. The features extracted by these networks are then used to build four different models: CNNAE-IVBRE, GCNAE-IVBRE, LSTMAE-IVBRE, and TransformerAE-IVBRE. These models are compared with the performance of the TFE-IVBRE model to conduct an ablation study for the TFE network. The performance metrics for each model on the DC and SRU testing datasets are presented in Table 4. The differences in prediction results across these methods reveal that when AE is built using CNN and GCN for TFE, the model's prediction performance degrades, which is likely due to their inability to capture long-term global dynamic relationships. When LSTM and Transformer AEs are used, the prediction performance of the model is closer to that of TFE-IVBRE. However, since these models fail to effectively capture local dependencies, there still remains a gap in prediction performance when compared to the TFE-IVBRE model, which captures both dynamic and local features using the HTFEAE network. The result comparison reflects the rationality of the HTFEAE network design.

Subsequently, the functionality of each component in the IVBRE model is examined. To be specific, the roles of HTFEAE feature extraction, the minor update mechanism in the incremental update process, the overall incremental update functionality, and the clustering-based ensemble learning approach are evaluated. To this end, the following models are constructed: the IVBRE model without TFE, the TFE-IVBRENM model disabling the minor update functionality, the TFE-VBRE model without incremental update capability, and the TFE-VBR model, which excludes the incremental update mechanism and is based solely on a single VBR model. Except for the removed components, all other parts of these models retain the same as the original TFE-IVBRE model. The performance metrics of these models on the DC and SRU testing datasets are shown in Table 5.

The ablation study results indicate that, on the DC dataset, the TFE capability of HTFEAE contributes the most to the performance of the TFE-IVBRE model. This finding aligns with the superior prediction performance of the models based on time series data observed in the comparative experiments, further emphasizing the importance of using time series data for soft sensor modeling. In addition, the online update ability of each VBR sub-model, or the minor update capability of the TFE-IVBRE, also plays a significant role in improving the prediction performance. When the entire incremental update functionality is removed, the model's prediction performance further declines, validating the effectiveness of the overall incremental update strategy. Finally, by comparing the testing results of the TFE-VBRE and TFE-VBR models, the positive impact of clustering-based ensemble learning on enhancing prediction performance is demonstrated. The results on the SRU dataset indicate that the absence of TFE still significantly harms the overall performance of the model. However, when the incremental update mechanism is removed, the negative impact on performance is even more pronounced. This is likely because the model's online update mechanism plays a more crucial role in maintaining soft sensors' performance during a prolonged operating time. This further highlights the effectiveness of the designed incremental update strategy.

4.7. Robustness test

Finally, the robustness of the TFE-IVBRE model is tested to verify whether it can maintain robust performance in scenarios where missing data and noise problems exist. First, data in the datasets are randomly

Table 5
Performance metrics of IVBRE ablation study methods on the DC and SRU testing datasets.

DC dataset					SRU dataset				
Model	MAE	RMSE	SPCC	R^2	Model	MAE	RMSE	SPCC	R^2
IVBRE	0.1079	0.1389	0.9531	0.5701	IVBRE	0.0186	0.0333	0.8800	0.6622
TFE-IVBRENM	0.0539	0.0750	0.9558	0.8748	TFE-IVBRENM	0.0136	0.0294	0.8581	0.7366
TFE-VBRE	0.0642	0.0944	0.9533	0.8017	TFE-VBRE	0.0223	0.0407	0.7278	0.4942
TFE-VBR	0.0921	0.1169	0.9509	0.6955	TFE-VBR	0.0251	0.0465	0.5638	0.3418
TFE-IVBRE	**0.0395**	**0.0461**	**0.9953**	**0.9513**	**TFE-IVBRE**	**0.0097**	**0.0210**	**0.9506**	**0.8652**

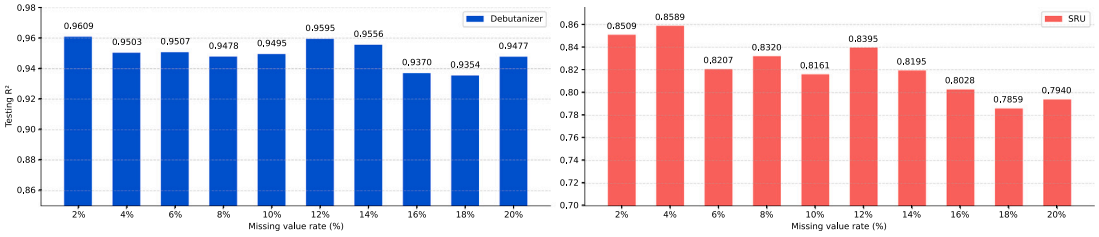


Fig. 13. TFE-IVBRE's missing value testing results.

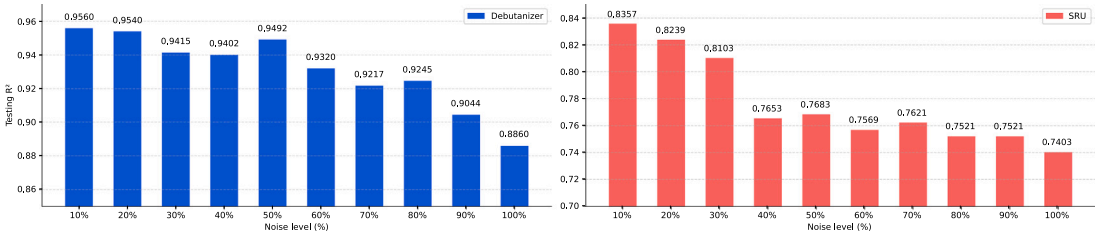


Fig. 14. TFE-IVBRE's noise testing results.

replaced with missing values to simulate sensor failures and test the TFE-IVBRE model's modeling and prediction performance under data missing conditions. The proportion of missing values ranges from 2% to 20%. For the missing data points, forward filling, a commonly used method in practical applications is employed for imputation. The TFE-IVBRE model is retrained and tested on datasets with varying missing value proportions, and the R^2 scores for each test are recorded, as shown in Fig. 13. According to the testing results in the DC and SRU scenarios, the TFE-IVBRE model exhibits outstanding robustness to missing values in the dataset. Even when the missing value ratio reaches 20%, the model still maintains good prediction performance. Its robustness on missing values can be attributed to the fact that, compared to using data snapshots, the TFE network benefits from utilizing data series. Even with a considerable amount of missing values in the series, the simple forward filling method can effectively compensate for this issue for data series construction, resulting in minimal information loss for the TFE-IVBRE model.

In another set of experiments, Gaussian noise is added to the dataset to simulate the commonly encountered poor data quality issue in industrial data and to test the performance of the TFE-IVBRE model on the DC and SRU dataset after noise has been added. The intensity of the added Gaussian noise is determined by its standard deviation, which ranges from 10% to 100% of the standard deviation of the original data. The performance of the TFE-IVBRE model on datasets with varying intensities of Gaussian noise is shown in Fig. 14. It can be observed that, as the noise intensity increases, the model's performance gradually declines. In the long-term operation of the SRU scenario, the performance degradation is reasonably more pronounced. Nevertheless, the degree of performance decline remains within an acceptable range, reflecting the adaptability of the TFE-IVBRE model to messy industrial data in real-world industrial scenarios.

5. Conclusion and future work

To enhance the online quality index prediction performance of soft sensors, this paper integrates the powerful temporal feature extraction capabilities of deep networks with the strengths of ensemble learning and variational Bayesian regression for real-time incremental updates. This leads to the development of the Temporal Feature Extraction and Incremental Variational Bayesian Regression Ensemble (TFE-IVBRE) framework, an efficient and versatile soft sensor modeling solution. Based on experiments conducted in the DC and SRU scenarios, the following three conclusions can be drawn:

1. By combining a deep neural network in the form of an autoencoder with a semi-supervised loss function, valuable temporal features can be extracted to support both the soft sensor modeling process and industrial quality index prediction, while also enhancing the robustness of the soft sensor model.
2. Compared to using a single regressor for soft sensor modeling, the clustering-based ensemble learning method first divides the dataset into different clusters based on their data patterns and then builds corresponding sub-models for each cluster to form the overall ensemble model. This ensemble approach can effectively distinguish between different operating conditions and handle complex data distributions in industrial settings, which improves the soft sensor's prediction performance and reliability across different industrial processes.
3. Combining minor updates for the VBR sub-models and major updates for the entire ensemble model, an incremental update strategy is proposed. This strategy leverages newly acquired data to continuously update the ensemble model, significantly enhancing the soft sensor's real-time performance in industrial data streams. It underscores the importance of regressors' online learning capabilities and demonstrates how the incremental

update strategy improves the performance of online industrial quality index prediction.

Building upon the above conclusions and the current work presented in this paper, several promising future research directions can be identified. First, for deep network-based temporal feature extraction, the current use of a single network structure may not fully accommodate the complex, multi-step nature of industrial processes. Future work could explore parallel and collaborative feature extraction methods using multiple networks. Second, the incremental update strategy for ensemble models offers many opportunities for further investigation. For example, it is crucial to balance the number of sub-models, computational resource consumption, and online prediction performance within industrial data streams. This, as is clearly evident, is a dynamic multi-objective optimization problem. Finally, as discussed in this paper, addressing the issues of missing values and high noise in industrial data could lead to research on probabilistic prediction techniques (Ge and Chen, 2019; Rutkowska et al., 2024), offering more targeted solutions to these challenges. Overall, the online industrial quality index prediction method based on deep temporal feature extraction and incremental ensemble modeling offers a prospective path for the future development of soft sensor modeling and holds many avenues for further research.

CRediT authorship contribution statement

Yuchen Zhao: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization, Data curation, Project administration. **Rui Yang:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation. **Jie Ying:** Writing – review & editing, Validation, Resources, Methodology, Investigation, Conceptualization. **Song Wang:** Resources, Supervision, Validation, Writing – review & editing. **Zhenyu Zeng:** Resources, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT based on Generative Pre-trained Transformer 4.0 (GPT-4.0) in order to improve the quality of the article writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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